H HireLens

AI Job Description Analyzer

From a raw job posting to a structured hiring report, using a 4-node LangGraph pipeline.

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Program: Artificial Intelligence and Data Science, SMIT

Problem and objective

The problem

Job postings are long and unstructured. Recruiters and candidates spend time hunting for the title, experience, skills and job type, and often miss what matters most.

The objective

Build an AI assistant that analyses a job description and converts it into a clear hiring report: title, company, experience, category, employment type, technical and soft skills, and the top 3 skills.

LangGraph workflow



Each node reads the shared state, adds its own result and passes the state to the next node.

Tech stack

Python

Core language for the whole backend

LangGraph

Defines the 4 nodes and passes state between them

Groq API

Fast LLM inference (Llama 3.3 70B)

FastAPI + Pydantic

REST API and strict data validation

HTML, CSS, JavaScript

Dark futuristic frontend, no framework

Firebase Auth + Vercel

Email/Google login and hosting

System architecture

Browser

HTML, CSS, JS UI + Firebase login

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FastAPI

/api/analyze and /api/match, input validation

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LangGraph

extract, classify, skills, report

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Groq LLM

Returns structured JSON per node

Empty or too-short input is rejected with a clear message before any LLM call is made.

Features

- Login with Email or Google (Firebase Auth), plus guest mode
- Hiring report with skill tags and top 3 skill score bars
- Resume match: score, matched skills, missing skills, advice
- Compare two job descriptions side by side
- History of past reports saved per user
- Export report as JSON or PDF

Also included

Animated particle background, loading steps that follow the pipeline, interview questions for recruiters, and graceful error messages for empty or invalid input.

Example output

Input

We are hiring an AI Engineer at ABC Technologies (full-time, on-site). You have 2-4 years of experience with Python, FastAPI, LangGraph and SQL. Strong communication, teamwork and problem solving are a must.

Hiring report

Job title: AI Engineer

Company: ABC Technologies

Experience: 2-4 years

Category: Artificial Intelligence

Employment: Full-time

Technical: Python, FastAPI, LangGraph, SQL

Soft: Communication, Teamwork, Problem solving

Top 3: Python, LangGraph, FastAPI

Frontend design

Dark futuristic theme

Deep navy background with cyan and violet glow, glass panels and an animated particle network.

Unique login

A pulsing 4-node pipeline shows the workflow before you sign in. Email, Google or guest access.

Responsive and accessible

Works on mobile, shows keyboard focus and respects reduced-motion settings.

Deployment

- Code on GitHub: api/ (FastAPI + LangGraph) and public/ (frontend)
- Vercel serves the frontend and runs the FastAPI function
- GROQ_API_KEY stored as a Vercel environment variable
- Firebase Authentication handles Email and Google sign-in
- Vercel domain added to Firebase authorized domains

Challenges and future scope

Challenges

Getting reliable structured JSON from the LLM, passing state cleanly between nodes, and handling empty or messy input without crashing.

Future scope

Upload PDF job descriptions, store history in Firestore, rank multiple resumes for one job, and add salary and market-trend insights.

Thank you

HireLens: read the posting, get the hiring report.

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